SEPARATEDNESS OF THE STRUCTURE MORPHISM IN THE VANISHING SITUATION, VIA AN INTEGRAL AFFINE SCHEME COVER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Existence of a scheme integral and surjective over the space. Decent Spaces, label decent-spaces-lemma-there-is-a-scheme-integral-over (invoked at proof line 3139). It supplies the scheme Y with a surjective integral morphism $Y \to X$, which is affine and therefore covered by the base change vanishing.
- Vanishing along an affine base change, and the affine criterion for a quasi-compact scheme. Same chapter spaces-cohomology.tex, label lemma-vanishing-base-change (invoked at proof line 3141) and Cohomology of Schemes, label coherent-lemma-quasi-compact-h1-zero-covering (invoked at proof line 3148). The first gives the vanishing of H^1 of quasi-coherent modules on Y; the second, since Y is a quasi-compact scheme, makes Y affine.
- An integral morphism is universally closed, and the image of a universally closed morphism is separated. Morphisms of Spaces, labels spaces-morphisms-lemma-integral-universally-closed (invoked at proof line 3152) and spaces-morphisms-lemma-image-universally-closed-separated (invoked at proof line 3155). They transfer the separatedness of the affine scheme Y to the algebraic space X.

Source reference. Original lemma 01XF: sources/stacks-chow/tags.txt, source lines 2516–2516. Navigation pointer for \ref{coherent-lemma-quasi-compact-h1-zero-covering} used inside the counted span (line 3149); the argument that consumes it is retained in the span.

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Source reference. Original lemma 09YB: sources/stacks-chow/tags.txt, source lines 12559–12559. Navigation pointer for \ref{decent-spaces-lemma-there-is-a-scheme-integral-over} used inside the counted span (line 3139); the argument that consumes it is retained in the span.

Source reference. Original lemma 07V0: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3041–3041. Navigation pointer for \ref{lemma-vanishing-base-change} used inside the counted span (line 3142); the argument that consumes it is retained in the span.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3133); the argument that consumes it is retained in the span.

Source reference. Original lemma 05Z2: sources/stacks-chow/tags.txt, source lines 7676–7676. Navigation pointer for \ref{spaces-morphisms-lemma-image-universally-closed-separated} used inside the counted span (line 3155); the argument that consumes it is retained in the span.

Source reference. Original lemma 0415: sources/stacks-chow/tags.txt, source lines 5280–5280. Navigation pointer for \ref{spaces-morphisms-lemma-integral-universally-closed} used inside the counted span (line 3153); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 07UY the morphism $p : X \to \mathrm{Spec}(A)$ is separated.*

Proof. By Decent Spaces, Lemma 09YB we can find a scheme Y and a surjective integral morphism $Y \to X$. Since an integral morphism is affine, we can apply Lemma 07V0 to see that $H^1(Y, \mathcal{G}) = 0$ for every quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$. Since $Y \to X$ is quasi-compact and X is quasi-compact, we see that Y is quasi-compact. Since Y is a scheme, we may apply Cohomology of Schemes, Lemma 01XF to see that Y is affine. Hence Y is separated. Note that an integral morphism is affine and universally closed, see Morphisms of Spaces, Lemma 0415. By Morphisms of Spaces, Lemma 05Z2 we see that X is a separated algebraic space. □